



Precision manipulation, adaptive navigation, and autonomous mobility for the advanced rover

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1 Introduction

Inspired by the recent success of ISRO's Chandrayaan-3, our team is excited to join the IROC-U 2024 competition and contribute possible technologies for near-future space exploration missions. It's a great opportunity for us to showcase our cool ideas for robotic rovers and work with ISRO on space technology.

Inspired by ISRO's Chandrayaan simplicity and efficiency in design, we have developed a rover that is both simple and highly efficient. Our chosen design incorporates a rocker-bogie mechanism with a differential linkage, optimizing the rover's mobility, especially in challenging M sand terrain. Opting for a four-wheeled configuration instead of six adds an element of control, particularly in intricate manoeuvres. For environmental awareness, we are equipping the rover with two cameras and an IMU to detect obstacles in its path. Additionally, we're employing LiDAR technology to identify craters effectively. The manipulator arm, constructed from scratch to adeptly pick up samples and deposit them into a designated container, holds a significant role. To ensure precise arm movements, we're implementing inverse kinematics. The integration of lightweight servo motors enhances balance and ease of control. Integration of multiple sensors for redundancy underscores our commitment to operational continuity and accuracy. This strategic approach not only safeguards against sensor failures but also enhances the overall reliability of our rover. With these components seamlessly working together, we are eager to assemble a remarkable rover poised for success in the upcoming competition. The rest of the document provides details about our system.

2 System Architecture

2.1 Design of System Architecture

This section presents a rover's system architecture, illustrating the interconnected sub-systems in Fig.1, their functionalities, and the flow of data and commands. The diagram also provides a comprehensive overview of how each essential component collaborates to achieve seamless navigation, perception, and manipulator control.

2.2 Essential Sub-systems in Architecture

- **Sensors:** Essential components such as cameras, IMU, and LiDAR capture the environment, providing crucial data for rover actions.

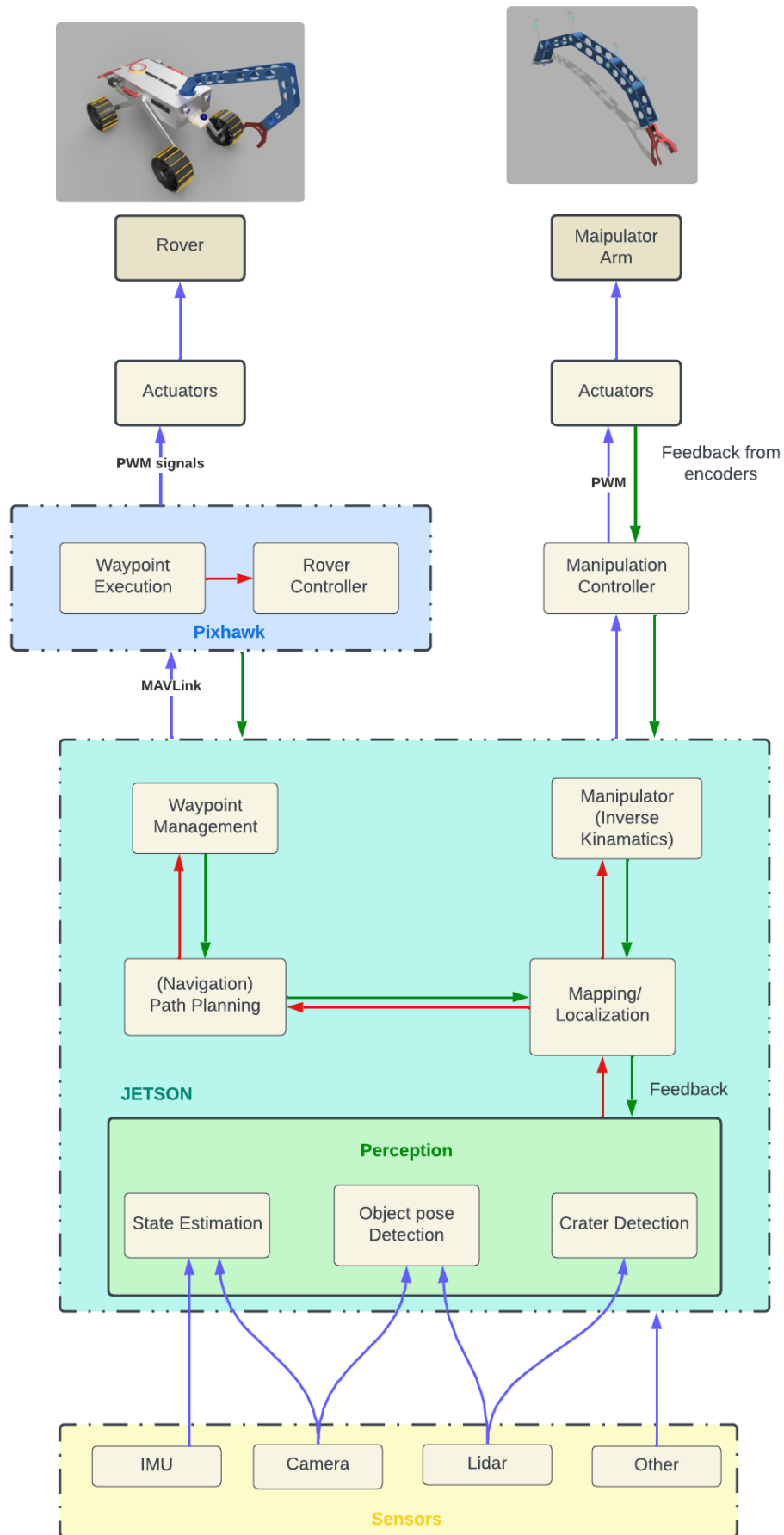


Figure 1: Rover setup and components

- **Perception:** This subsystem utilizes information from sensors to calculate object positions within the environment.
- **Mapping and Localization:** Based on perception, this subsystem determines the rover's position in the environment and the relative positions of objects.
- **Path Planning:** Necessary for navigating the rover through environments filled with obstacles and craters, this subsystem chooses optimal paths between positions.
- **Waypoint Management:** Maintains a list of previously visited and upcoming rover positions, facilitating effective path tracking.
- **Waypoint Execution:** Implements given waypoints, ensuring the rover follows the designated path in action.
- **Rover Controller:** Drives rover movements based on gathered information, providing control for effective navigation.
- **Manipulator Inverse Kinematics:** After obtaining sample information, calculates angles to facilitate arm movement for pick and drop actions.
- **Manipulator Controller:** Utilizes calculated angles to drive arm movements, ensuring effective execution of pick and drop tasks.

2.3 Interfaces among different Sub-systems

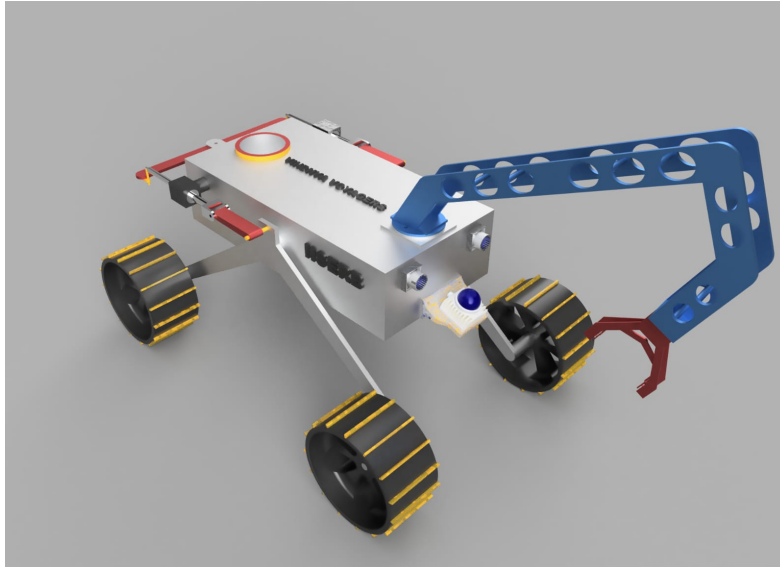
- **Sensors to Perception:** Raw data from sensors, including cameras, LiDAR, and IMU, will be transmitted to the perception subsystem. Utilizing information from cameras and IMU, along with provided initial position coordinates, state estimation for object and crater poses will be calculated.
- **Perception to Mapping and Localization:** The perception module will determine the estimated poses of objects and craters. This information will be forwarded to the Mapping and Localization subsystem, where a local map of the environment will be generated to support subsequent processes.
- **Mapping and Localization to Path Planning:** The localized mapping coordinates will be communicated to the Path Planning subsystem. Algorithms for navigation, such as calculating the shortest path, object traversal, and object detection, will be applied in this phase.
- **Path Planning to Waypoint Management:** After path planning, the identified points for rover movement will be passed to the Waypoint Manage-

ment subsystem. Waypoints for navigating through the environment will be maintained in this interface.

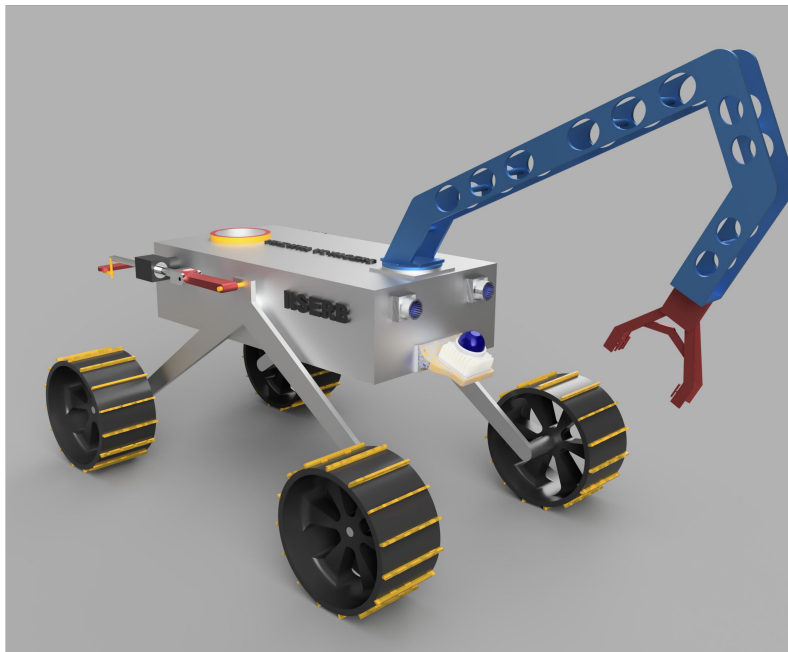
- **Waypoint Management to Waypoint Execution:** Waypoints for the current movement will be transmitted to Pixhawk in this interface, managing the sequential execution of waypoints to guide the rover.
- **Waypoint Execution to Rover Controller:** Pixhawk, based on received waypoints, will issue instructions to the motor drivers through this interface, governing the movement of the rover.
- **Rover Controller to Actuators:** The Rover Controller directs rover movement by sending precise commands to the actuators. Integrated encoders provide real-time feedback, confirming the successful execution of movements. This bidirectional communication ensures accurate control and verification of the rover's actual position.
- **Mapping to Manipulator:** Jetson, based on sample information, will provide the sample's coordinates. These coordinates will undergo processing using inverse kinematics to calculate the angles for each joint in the manipulator.
- **Manipulator(Inverse Kinematics) to Manipulator Controller:** Derived angles from inverse kinematics are transmitted to the Manipulator Controller in Pixhawk, directing precise movements of the manipulator's links. Simultaneously, feedback from the controller refines and verifies the manipulator's positions, ensuring accurate and responsive operations.
- **Manipulator Controller to Manipulator Actuator:** Manipulator Controller transmits calculated instructions to the motor drivers of the manipulator's arm, directing precise movements as per the specified parameters. Concurrently, feedback from encoders on the actuators is sent to the Manipulator Controller, facilitating continuous monitoring and adjustment for accurate control and positioning.”

3 Roving mechanism

3.1 Schematic Representation



(a) Isometric View



(b) Diametric View

Figure 2: Different view of the 3D CAD model of the Rover

3.2 Roving Mechanism

The rover incorporates a 4-wheel rocker suspension system [19] with a differential bar linkage[10], enabling our rover to maintain contact with surfaces[5] on uneven terrains. For manoeuvring, we are using a differential drive, which ensures precise direction control by adjusting the speeds of the four independently driven wheels[4][15]. To enhance traction, we employ custom-designed 3D-printed tyre grips. The rocker mechanism connects forward and backward tires on both sides, rotating freely around an anchored point on the chassis. Connecting the rocker mechanisms of both sides through a differential linkage stabilizes the four-wheel chassis, providing passive suspension and stability, which is especially beneficial for overcoming obstacles. Furthermore, the independent power of each wheel not only facilitates efficient manoeuvring through challenging terrains but also contributes to the overall redundancy of the rover's propulsion system. This redundancy ensures that the rover can continue its mission even if one or more wheel actuators encounter issues, promoting reliability in diverse operational scenarios.

3.3 Explanation for Selecting the Proposed Mechanism:

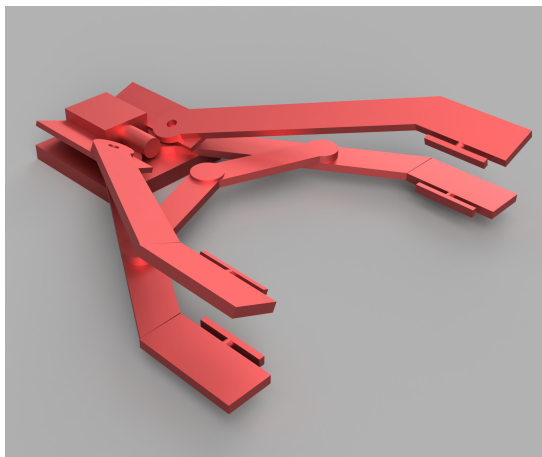
- **Rocker mechanism:** The 4-wheel rocker design with differential linkage excels in terrain adaptability by maintaining passive contact with all wheels on uneven surfaces, optimizing traction and stability. In contrast, a normal 4-wheel design may result in loss of contact for other wheels when one wheel rises due to the rigid body, limiting adaptability on uneven terrains. The rocker design overcomes this limitation, ensuring continuous contact for enhanced performance. Additionally, differential linkage provides stability by limiting the chassis's angular movement to half the degree of the rocker mechanism. This arrangement ensures that the rover's chassis remains relatively steady, even when the rocker mechanism allows for greater flexibility in the movement of individual wheels.
- **Four-Wheel Advantage:** The 4-wheel rocker mechanism holds a distinct advantage over the 6-wheel configuration with its key strengths in reduced complexity, improved operational efficiency, and fewer points of failure. The simplified design, achieved by employing fewer components, translates to easier maintenance and heightened operational efficiency. This streamlined approach not only enhances the rover's responsiveness across diverse terrains but also minimizes the potential for mechanical issues.

- **Skid steering:** It offers precise and agile manoeuvring, simplicity, reliability, smooth transitions between forward and reverse motion, and better adaptation to uneven terrain compared to traditional steering mechanisms, as we can individually control the speed of each wheel.
- **Custom tyre grip:** Custom-designed 3D-printed wheel grips provide tailored traction, adaptability to mission needs, lightweight durability, and rapid prototyping for optimized rover performance in diverse environments.

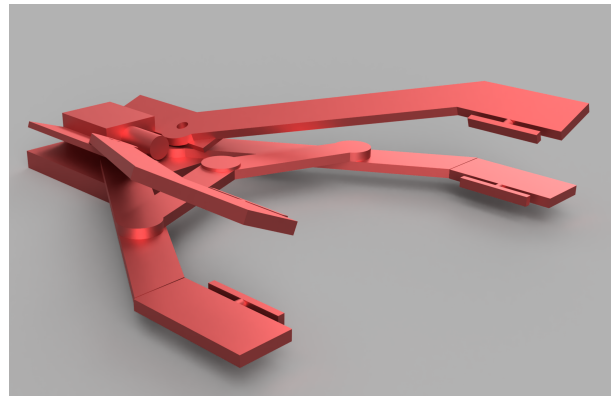
The selected mechanism aligns with our mission objectives by prioritizing simplicity, weight management, energy efficiency, and adaptability to diverse terrains. This choice's comprehensive understanding and justification underscore its suitability for robust and effective rover exploration.

4 Mechanism for Sample Pick-and-Place Activity

4.1 Schematic Representation



(a) isometric view



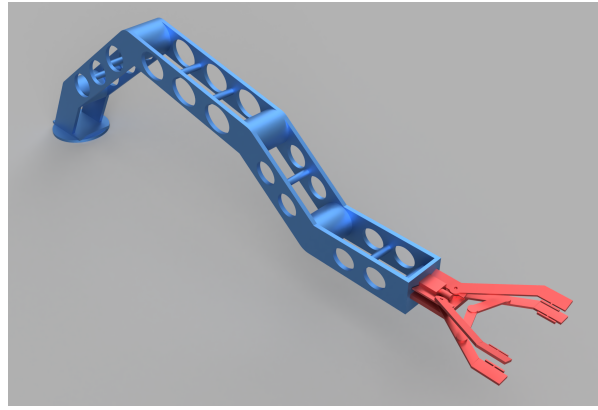
(b) Dimetric View

Figure 3: Tentative Design of Claw for Manipulator [only isometric and dimetric views](#)

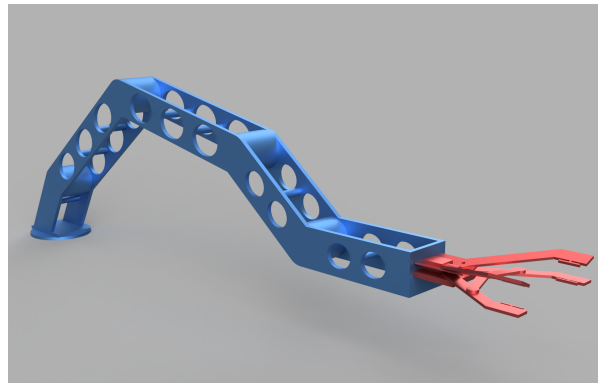
4.2 Schematic Representation

A precision-oriented and consistent approach to sample pick-and-place tasks is achieved through the implementation of a 5-degree-of-freedom (5-DOF) + gripper manipulator arm. Grubler's formula was utilized to ascertain the degrees of freedom [11, 17].

The electric gripper is used as a manipulator's arm end-effector, enhancing control over gripping force, position, and speed. **A camera is strategically positioned on the gripper**, working in tandem with rover cameras to detect



(a) isometric view



(b) Dimetric View

Figure 4: Tentative Design of Claw for Manipulator [only isometric and dimetric views](#)

the pose of sample tubes and their containers. The gripper is equipped with a force-sensing resistor for grasping force detection and an IR sensor to measure the distance from the gripper to the sample tube. This configuration ensures a precise and responsive handling of samples.

The manipulator arm's design is optimized for a 3D workspace of at most 0.7 m^3 , determined using Robotics Toolbox for Python [2], making it well-suited for the specified task.

Materials such as acrylic and ABS plastic were selected for the manipulator arm due to their lightweight nature, cost-effectiveness, and corrosion resistance. The arm's dimensions, with a horizontal reach of 65-75 cm and a gripper effective length of 10-15 cm, were carefully chosen to align with the required 3D workspace and sample tube dimensions.

Servo motors with encoders were chosen for joint control, prioritizing low weight, compact design, high torque at low speed, and precise angle feedback using a PID controller. This selection ensures efficient and accurate movement and rotation of each joint. These motors are connected in parallel to ensure independent control, redundancy, and uniform torque distribution.

While an Inertial Measurement Unit (IMU) was initially considered for end effector position determination, it was later excluded due to the recognition of potential inaccuracies arising from integration drift associated with utilizing acceleration data [1]. Instead, a dedicated control algorithm will be developed, as outlined in section 4.3, to execute the manipulator arm's precise and consistent sample pick-and-place mechanism.

4.3 Mechanism description

The proposed arm is an autonomously operated system. The motion sequences are as follows:

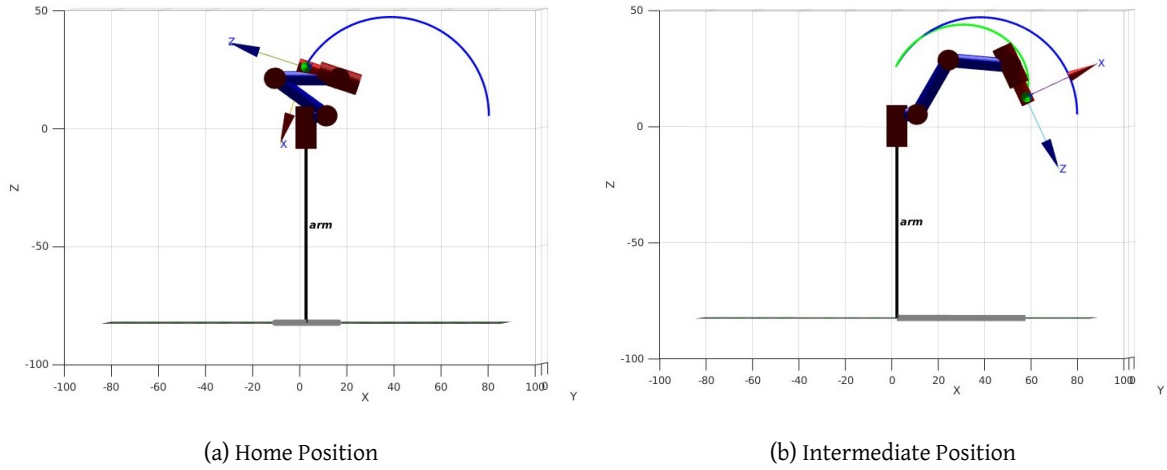


Figure 5: Different Positions of Manipulator arm

4.3.1 At Waypoint (WP):

1. Initially, the arm is at its home position as shown in Fig.5a.
2. The coordinates and the pose of the sample tube are given by two cameras attached in front of the rover and the camera on the manipulator arm as given in point 3b, section 8.3.
3. Once the coordinates and pose are received, the joint variables are determined in terms of the desired position of the end-effector using inverse kinematics.
4. After adjusting the angles, the end effector approaches the sample tube and stops at a distance of 2cm to 5 cm from the sample tube. This distance is calculated by the IR sensor attached to the end effector. The gripper

opens up and rotates about its axis so that the fingers of the end effector are perpendicular to the longitudinal axis of the sample tube.

5. When the sample is between the fingers, it starts to close. Force-sensing resistors on the fingers determine the force with which the gripper will hold the sample.
6. The manipulator now goes to a predefined intermediate position as shown in Fig.5b before rotating around its base at an angle of 180 degrees. The angle of the joints is set depending on the placement of the container (so that the arm can go down to put the sample and up in case of picking up the sample). The ArUco markers [6] surrounding the container are detected by the camera on the end-effector. The sample tube is placed in the container. Here the container is a gripping mechanism where when the sample tube is placed a force sensor detects it and the gripper grasp it. Similarly while picking the sample tube back the manipulator first hold the tube then the gripper releases it.
7. The manipulator arm returns to its home position (Fig.5a) via intermediate position (Fig.5b).

4.3.2 At Final point (FP):

1. The arm starts at its home position, goes to its predefined intermediate position (Fig.5b) and rotates 180 degrees around its base.
2. The camera attached to the manipulator's arm detects ArUco markers surrounding the container to determine the placement of the container on the rover. The camera detects the pose of sample tube similar to point 3b, section 8.3.
3. The steps 3 to 5 as mentioned in section 4.3.1 which are concerned with inverse kinematics calculations, adjusting angles of joints and gripper movement given above are repeated.
4. The base of the arm rotates around at an angle of 180 degrees to reach its intermediate position (Fig.5b).
5. The cameras on the rover and the manipulator arm detect the container at the FP and give its coordinates.
6. The joint variables are determined in terms of the desired position of the end-effector using inverse kinematics.
7. The end effector releases the sample inside the container. The force-sensing resistor detects the release, and its correct position is detected by the

rover's two cameras and the manipulator arm's camera.

The PID controller ~~continuously monitors and adjusts the angle between the links to maintain it as close as possible to the desired value, ensuring precise control within the system.~~

4.4 Justification for Selecting the Proposed Mechanism:

- **Predefined Home Position:** Returning to the home position between manipulations ensures a predictable starting point for each action, maximizing efficiency and maintaining the centre of gravity of the rover while the manipulator is not being used.
- **Predefined Intermediate Position:** This serves as a collision-free staging area for pre-manipulation and post-manipulation manoeuvres.
- **ArUco Markers:** These markers provide a readily detectable reference for locating the container on the rover and ensuring accurate placement of the sample.
- **Force Sensing Resistor:** It provides closed-loop control of the gripper, preventing excess force and ensuring optimal sample tube integrity.
- **IR sensor:** The IR sensor mitigates collision risks through continuous distance measurement, optimizing gripper trajectory and grasp execution.

The proposed manipulator arm demonstrates a meticulous fusion of design elements, material choices, and operational features. It is an ideal, effective solution for autonomous sample handling within the rover system.

5 Emergency Response System

In the face of unexpected events, we have considered potential failures encompassing both hardware and software aspects. Our comprehensive approach involves addressing hardware failures, such as those in the rover's mechanical components, as well as potential software glitches that might impact the rover's computational capabilities. To overcome these challenges, we have devised strategic plans, including hardware redundancy and adaptive software protocols. These measures aim to ensure the rover's resilience and uninterrupted functionality by swiftly responding to any unexpected occurrences, thereby enhancing its overall robustness and reliability.

5.1 LiDAR Failure

- If the LiDAR fails, the rover will use the stereo cameras for crater detection.

5.2 Camera Failure

- **Onboard Camera:** If one of the cameras fails, the rover will switch to monocular VIO. If both cameras fail, the rover switches to VIO using the manipulator camera. If the manipulator camera also fails, the rover switches to Li-DAR based SLAM.
- **Manipulator Camera, While picking up the sample:** In the scenario of picking up a sample from the ground, the manipulator arm's functionality is entirely reliant on the rover cameras. When picking up a sample from the rover's container, the manipulator arm follows a predetermined joint orientation established by the rover. Additionally, the integration of an IR sensor and force-sensing resistors comes into play to compensate for variations. These sensors collaboratively work to detect the distance of the sample tube and ascertain the optimal force required for secure sample tube manipulation.
- **Manipulator Camera, While placing the sample in container:** When tasked with placing the sample tube in the rover's container, the manipulator's arm adheres to predetermined joint orientations specifically designed to facilitate seamless access to the container. When the manipulator arm is assigned the task of placing the sample tube in the container at the final point, its reliance is solely placed on the rover cameras for guidance. The IR sensor and force-sensing resistor play a crucial role in this process, detecting the distance to the container and determining the optimal force required to handle the sample tube effectively.

5.3 Sensor Failure

- **IR sensor:** In the event of a failure where the IR sensor on the gripper of the manipulator's arm ceases to respond, both the cameras on the rover and manipulator will actively attempt to compensate for this setback.
- **Force sensing resistor:** Should the force sensing resistor on the gripper of the manipulator arm cease to respond, the cameras on both the rover and manipulator arm will proactively attempt to compensate for this setback.

5.4 IMU Failure

- In the event of a malfunction in the external Inertial Measurement Unit (IMU), the system seamlessly transitions to utilizing the IMU embedded within the Pixhawk for Visual-Inertial Odometry (VIO). Should both the external and internal IMUs encounter failures, the rover's localization process is seamlessly shifted to rely on a combination of visual ophthalmology and Lidar data.

5.5 Battery Failure

- We're arranging the batteries in a series, so if one battery fails, it only reduces the voltage without affecting the entire system.

5.6 Motor Failure

- **Wheel motor failure** In the scenario of a motor failure in the rover, a pre-emptive response is implemented by swiftly disconnecting power from the affected motor. This strategic disengagement enables the rover to continue its operation using the remaining functional wheels, ensuring sustained mobility and functionality.
- **In Manipulator Arm, During task:** In the event of a motor failure, the rover will autonomously readjust its position to facilitate the manipulator arm's reach to the sample tube. Post-task completion, the operational motors will initiate an adjustment to rotate the manipulator arm back towards the rover, ensuring a return to the desired orientation.
- **In Manipulator Arm, After task:** In the event of a manipulator failure occurring after depositing the sample into the rover's container, the system will initiate a corrective procedure. The manipulator arm will actively attempt to rotate back toward the rover. Should a malfunction occur in the manipulator after successfully placing the sample tube in the container at the designated final point, the manipulator arm will initiate an automated response. It will actively attempt to rotate back towards the rover despite the completion of tasks.

5.7 A wheel breaks down

- If there is a failure in one or more than one wheel, the rover will try to move till the rover is immobilised due to the damaged wheel

Serial No.	Hardware Details	Category	Qty. No	Justification
01	DC geared motor	DC Motor	4	High torque, simple to control, high efficiency
02	DC Motor Driver	Motor Controller	2	Dual channel, bidirectional, buck-boost converter
03	11.1v 8AH Li-ion battery	Batteries	3	High energy density, lightweight
04	3D printing filament (TPU /PLA /Nylon, etc.)	Wheels	4	Durable, Flexible
05	Pixhawk	Flight control	1	open-source autopilot system, provides a platform for the control of autonomous vehicles
06	Aluminium Extrusions and Sheets	Chassis	As per requirement	Cost-effectiveness, Resistance to corrosion, High strength-to-weight ratio and easy to assemble
07	Steel angles/ bars/ rods	Differential linkage	As per requirement	Durable, low cost, Easy to use
08	Part of chassis	Acrylic Sheets	As per requirement	durable, easy to utilise, room for trial and error
09	Sample Container	??	??	??
10	Miscellaneous	Tools and Consumables	As per requirement	

Table 1: Hardware Identification for Roving Mechanism

5.8 Kill switch will be pressed

- When the rover goes out of the arena
- If the rover falls
- If there is smoke coming out of the rover

6 Hardware Identification

For enhanced clarity, we categorize the hardware identification table based on subsystems, namely the rover, manipulator, and navigation.

6.1 Hardware Identification for Rover

The [table 1](#) outlines the hardware identification for the **Roving Mechanism Subsystem**, detailing the necessary components required to construct the rover.

6.2 Hardware Identification for Manipulator

In the [table 2](#), we've included a complete list of all the materials and sensors essential for constructing the **Manipulator**, which is our next subsystem.

Serial No.	Hardware Details	Category	Qty. No	Justification
01	Actuators	Servo Motor	4	lightweight, ease of integration, precise feedback.
02	Actuators	Stepper Motor	1	holding position, holding torque, Positional accuracy.
03	Camera	720p USB Camera	2	lightweight, ease of use, better image processing and feedback, position estimate.
04	IR sensor	sensor	2	Compact size, lower power consumption, real-time feedback
05	Force sensing resistor	Force Sensor	2	Dimension-wise compatible, flexible, precise.
06	Servo and Stepper Motor Driver	Driver board	3	Can connect to 18 servos at a time independently
07	Acrylic Sheets	Chassis of the manipulator	As per requirement	durable, easy to utilise, room for trial and error
08	Battery	power source	1	6k to 10k mAh battery with output voltage 15V
09	Wires and Cables	Tools and Consumables	As per requirement	Assembly Necessities
10	Miscellaneous	Tools and Consumables	As per requirement	Assembly Necessities

Table 2: Hardware Identification for Manipulator

6.3 Hardware Identification for Navigation

In the table 3, we've included the essential hardware components required for the sensing and control aspects within the **navigation subsystem**.

7 Software Identification

7.1 Software Requirements

7.2 Algorithms

divide your subsection clearly

- HybVIO [13]

It is a hybrid approach for combining filtering-based visual-inertial odometry (VIO) with optimization-based SLAM. Highly robust, independent VIO with improved IMU bias modelling, outlier rejection, stationarity detection, and feature track selection, which is **adjustable to run on embedded hardware**.

Hybrid VIO algorithms combine the outputs from both cameras and IMUs to overcome their individual limitations. Cameras excel at providing accurate relative motion over short distances but suffer from accumulated

Serial No.	Hardware Details	Category	Qty. No	Justification
01	MID-360 LIVOX	LiDAR	1	Horizontal FOV up to 360° and Vertical FOV up to 59°.
02	FLIR Firefly@ S	Camera	2	Access to global shutter CMOS sensors, Ideal for embedding into compact portable devices
03	VectorNav VN-100	IMU	1	Provides calibrated IMU data, and it's continuous over the complete 360 degrees of motion. High data outputs: 400 Hz Navigation data and 800 Hz IMU data
04	NVIDIA@ Jetson AGX Xavier	On-board computer	1	Onboard GPU, low power consumption

Table 3: Hardware Identification for Navigation

drift over time and sensitivity to lighting conditions. IMUs can compensate for drift and offer continuous motion estimation but have inherent noise and accumulate errors rapidly without visual updates.

By fusing the data, hybrid VIO systems achieve:

- Improved accuracy and reduced drift compared to pure visual odometry.
- Improved robustness to visual challenges like darkness or motion blur.
- Continuous motion estimation even when the camera is temporarily unable to see.

Advantages

This algorithm beat the previous leader BASALT, on the EuRoC MAV benchmark. It also performs with high accuracy on the SenseTime-VISLAM dataset, which has occlusions. Tracks accurately even when the camera is covered.

• DOPE [16]

Deep Object Pose Estimation is a technique that uses Deep Learning to estimate the pose of 3D objects in the clutter from a single RGB image in real-time. DOPE addresses the problem of the reality gap in Deep Learning by generating synthetic pre-labelled data using domain randomization for training the model. In our project, DOPE is used to estimate the position and orientation of the sample tube and locate the cubical obstacles.

Method

1. **Belief Map Generation** A deep neural network generates belief maps

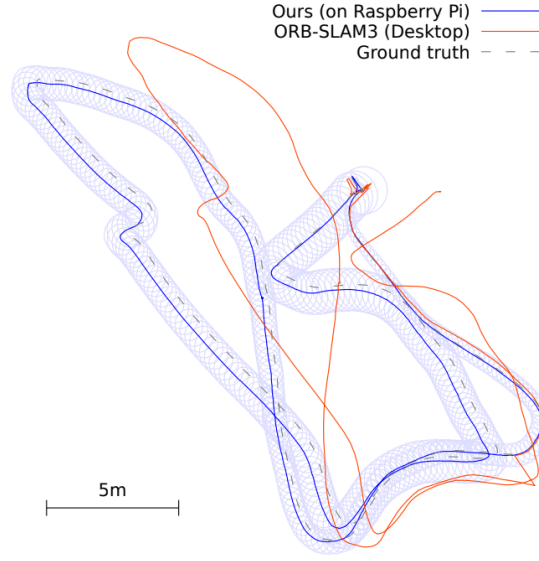


Figure 6: HybVIO vs. ORB-SLAM3 vs. Ground Truth

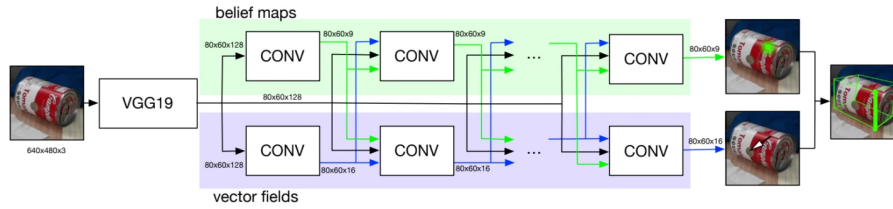


Figure 7: Network Architecture for DOPE

of 2D key points of all the objects in the image coordinate system. (centroid and corner points, i.e. features)

2. **6 DoF Pose Estimation** Identified peak locations from these belief maps are utilized as input to a standard perspective-n-point (PnP) algorithm for estimating the 6-degrees-of-freedom (6-DoF) pose of each object instance.

Implementation of DOPE for stereo-camera setup

1. The DOPE Algorithm requires only 1 camera for the purpose of 3-D pose estimation. However, we will be getting two point of views of the same object due to the presence of two cameras. By integrating the views of the two cameras, a larger field of view will help improve accuracy while picking up the sample as well.
2. Integrating the views from two cameras into a single representation for object pose estimation can be achieved through triangulation [8], which involves finding the intersection point of two or more lines of sight.

3. The following steps will be followed:
 - (a) **Calibrate Cameras:** Determine the intrinsic (focal length, principal point, lens distortion) and extrinsic parameters (position, orientation relative to each other).
 - (b) **Extract Features:** Identify the recognizable features in images captured by each camera.
 - (c) **Match Correspondences:** Establish correspondences between features in images from both cameras.
 - (d) **Triangulate for 3D Position:** Use the correspondences to triangulate and find 3D position of the object based on 2D feature positions in each camera's view.
4. Additionally, if one camera detects multiple similar objects that are not visible to the other camera, we aim to prevent false detections. To address this, we will assess the Intersection Over Union (IoU) of neighbouring objects detected by both cameras. A specific threshold for IoU will be established, and if met, the objects will be considered identical. This approach ensures more accurate detection by mitigating the impact of similar objects detected in one camera's view but not visible to the other.

$$\text{Intersection Over Union}(IoU) = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (1)$$

Advantages

- **Crater Detection Algorithm [20]**

The algorithm will detect the size and distance of the crater with the help of data obtained from the LiDAR. This algorithm uses a new method that uses 3-dimensional data to "fill in the gaps" of missing crater parts, aiming for better detection accuracy. The success of this method is measured by two factors: how well it completes the missing crater parts and how accurately it identifies areas that don't contain craters.

- **Navigation**

1. Start localization and mapping at the start point (0, 0) by initializing HybVIO and use DOPE and crater detection using LiDAR to create a map. The map is iteratively updated with the discovered obstacles and craters when the rover moves.
2. **Segment 1 (Commanded Waypoint Navigation)**

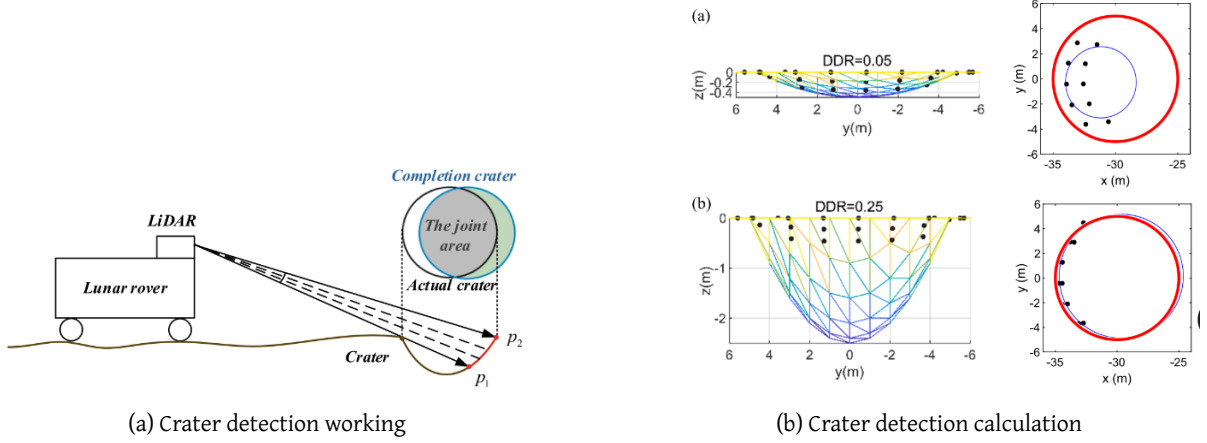


Figure 8: Crater detection using LiDAR

- (a) The coordinates of the next obstacle to be traversed is set to be the current goal G_i
- (b) A path planning algorithm such as A* [7], RRT [18], Dijkstra [3], D* [14] using the Open Motion Planning Library [9] is then used to generate a path from the current point to G_i and determine the next grid cell to go to and the coordinates of that grid are set to be the local goal g_i
- (c) After the rover reaches g_i , the path planning is run again, using the updated map and the next grid to travel, generating a new path and thus determining the next grid cell to travel to.
- (d) Steps 2a to 2c are repeated till the rover traverses G_i
- (e) Steps 2a to 2d are repeated for all the obstacles (G_i) required to be traversed in Segment 1
- (f) After traversing all obstacles and craters, G_i is set to be the way-point (WP in the arena diagram), and the rover then navigates to WP in the same process followed above.

3. Segment 2 (Autonomous Sample Pickup)

- (a) After reaching the waypoint WP, the rover approaches the sample tube till it reaches a distance of 35 cm from the sample.
- (b) DOPE is run to get the sample tube's pose (coordinates and orientation), which is communicated to the manipulator arm for picking up the sample. Refer Section 4.3.1

4. Segment 3 (Autonomous Waypoint Navigation)

- (a) The further navigation will be fully autonomous.
- (b) The final container position will be detected by the manipulator camera by scanning the arena using DOPE
- (c) This position is set as the end goal position for the path planning algorithm.
- (d) The rover then travels to the final container position in a similar way as described in Segment 1 by setting G_i to the final point and iteratively updating the path based on the obstacles detected during this manoeuvre.

8 Hardware and Software Realization Plan

8.1 Table for Hardware and Software Realization Plan

8.2 Realization Plans for Fabricated Hardware:

8.2.1 Rover:

Section outlines the realization plan for the fabrication of a rover. The objective is to detail the step-by-step process involved in transforming project requirements into a tangible, functional rover.

- **Dimension and Specification Finalization:**
 - Establishing key details based on project requirements.
- **Hardware Component Procurement:**
 - Initiating the acquisition of necessary hardware components.
- **Fabrication Process:**
 - Commencing the fabrication process with a focus on designing and creating the rover chassis, incorporating the rocker mechanism and differential linkage.
- **Preliminary Testing:**
 - Conducting initial tests to ensure functionality and stability of fabricated chassis.
- **Basic Motion Testing:**
 - Connecting wheels, battery, motor controller, and motors to test basic

motion using a remote control and Pixhawk for mobility verification.

- **Navigation System Integration:**

- Mounting and calibrating a stereo camera and LiDAR sensors as per navigation team requirements.

- **Algorithm Implementation:**

- Implementing algorithms for autonomous straight-line movement and obstacle avoidance using stereo camera and Jetson.

- **Manipulator Arm Addition:**

- Introducing a manipulator arm to the rover chassis, focusing on assessing stability during various arm motions.
- Adding the sample holding mechanism and testing its grip.

- **System Integration and Testing:**

- Ensuring seamless interaction and functionality of all integrated systems.

- **Performance Optimization and Refinement:**

- Fine-tuning algorithms, parameters, and control systems for optimal rover functionality.

8.2.2 Manipulator:

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1. **Pre-fabrication:**

Simulation: A comprehensive simulation of the entire system, including the manipulator arm mounted on the rover, will be conducted.

2. **Post-fabrication:**

Manual control using RF: The movement of the arm will be carefully assessed to identify any issues, such as jerky or vibrating motions. These concerns may be attributed to factors like the weight and torsion of the arm, motor speed, acceleration, and friction. Accordingly, adjustments to parameters like arm length may be made or motors may be changed, and inverse kinematics calculations will be performed based on the modified parameters.

Analysis of gripper movement: Data from the force-sensing resistor on the gripper will be analyzed to determine the force required for the

safe handling of the sample tube. Simultaneously, data from the IR sensor will be examined to understand the appropriate distance needed to reach the sample tube. Necessary modifications to the gripper design will be implemented based on these analyses.

Development of control algorithm: The integration of Python and the Robot Operating System (ROS) will be employed for autonomous arm control. This will be followed by the development and implementation of an effective and efficient control algorithm to enhance the overall performance of the system.

8.3 Implementation strategies for software

The following will be the sequential steps to implement the software architecture and test all fail-safe mechanism:

1. Evaluation involves rigorous testing of the HybVIO, DOPE, and Crater Detection algorithms using open-source datasets in an offline mode.
2. Integration with ROS2 is implemented to facilitate seamless functionality and interaction between the algorithms.
3. A simulation world is meticulously designed in Gazebo, resembling the physical arena for subsequent testing and analysis.
4. The URDF model of the rover is launched in Gazebo, featuring camera, LiDAR, and IMU placements to replicate real-world conditions.
5. Simultaneous testing of all algorithms within the Gazebo simulation is conducted, and the accuracy of each algorithm is thoroughly analyzed for performance assessment.
6. Fail-safe mechanisms are rigorously tested within the Gazebo simulation environment to ensure their effectiveness in handling potential issues.
7. Communication tests are conducted between different components of the real rover, including Jetson Xavier and Pixhawk, Pixhawk and motor drivers, and sensors and Jetson.
8. The algorithms are deployed on the physical rover, and a thorough analysis of their accuracy is conducted to validate their performance in real-world scenarios.
9. Fail-safe mechanisms are tested comprehensively on the physical rover to ensure their robust and reliable performance in real-world conditions.

9 Test Plan

The objective of this test plan is to comprehensively evaluate the rover's performance under various configurations by systematically testing different permutations of the rover, navigator, and manipulator setups. The testing will involve systematically combining different configurations of the rover, navigator, and manipulator components to assess their individual and collective impact on the overall system performance. This approach aims to identify any potential issues, evaluate the robustness of the system, and ensure seamless integration across various setups. The test scenarios will include permutations of the following configurations:

9.0.1 Rover

- **Terrain Testing:** Flat surfaces, uneven terrains with obstacles, inclined surfaces.
- **Wheel Speed Variation:** Gradual acceleration and deceleration, constant low-speed operation, constant high-speed operation, and sudden speed changes to simulate dynamic environments.
- **Mechanism Testing:** Differential steering on flat surfaces, manoeuvring through tight spaces, assessing responsiveness to abrupt changes in direction, simulating turning on inclined surfaces.
- **Inclination Testing:** Ascending and descending moderate inclines, ascending and descending steep inclines, side-hill operations to assess lateral stability, and dynamic incline changes during motion.

9.0.2 Navigator

- **Path Planning Algorithm Testing:** A* algorithm, Dijkstra's algorithm, RRT algorithm, etc.
- **Sensor Configuration Testing:** Navigation using only onboard cameras, manipulator-mounted camera, Fusion of data from both onboard and manipulator cameras, LiDAR-only navigation, Combination of lidar and camera data for improved perception.
- **Camera-LiDAR Combination Testing:** Only with cameras, Only with LiDAR, one camera and one LiDAR, etc.
- **Environmental Variation Testing:** Navigation in well-lit environments, Navigation in low-light or dark conditions, Changes in ambient lighting

during navigation, The presence of shadows and reflections, and Navigation in environments with dynamic lighting conditions.

- **Obstacle Position Variation Testing:** Obstacle-free navigation, Fixed-position obstacles, obstacles with changing positions, and Obstacles at varying distances and sizes.

9.0.3 Manipulator

- **With and without Camera in Manipulator:** Using information from a camera on the manipulator and onboard cameras, only using onboard camera information
- **Different Positions:** Camera at the tip of the manipulator, camera on the side of the manipulator, camera on the base of the manipulator, evaluate the impact of camera position on visibility and manipulation accuracy.
- **ArUco Marker Recognition:** with ArUco marker in the rover(one marker, multiple markers), without markers
- **Container Position Variation:** Container positioned at different sides of the rover based on manipulator position.

10 System Specification

11 Project Management

~~Utilizing Notion as our task management platform~~, we have organized a task list that aligns with the expertise, past experience, and specific areas of interest of each team member. Tasks have been strategically divided to capitalize on individual strengths and interests. For each task, we have provided tentative cost and estimated time, which are detailed in Table 10.

12 Novelty

- **Handling Drift in Localization** The rover incorporates HybVIO which uses *two cameras* along with an IMU to perform localization, which helps mitigate the effect of IMU drift because of the use of vision. Also, there is no possibility of error due to slipping of wheels as wheel motor-encoder based odometry is not being used.
- We have coded a robust crater detection system using LiDAR data, making

our rover adept at navigating challenging terrains and identifying potential hazards.

- **Final Goal Detection** The *manipulator* camera scans the area and uses DOPE to determine the location of C the final container, improving path generation time by reducing search space size.
- **Localization, Mapping and Path Planning** Localization is being done using state-of-the-art algorithms like HybVIO [13] and robust and accurate algorithms like DOPE [16] and LiDAR based crater detection are used for mapping. All the navigation computation is performed on powerful machines like Jetson Xavier and Pixhawk
- **Crafting our rover involves a hands-on approach, creating custom components like wheels, manipulator frames, and chassis. This not only shows our dedication to customization but also ensures each part meets our specific project needs.**
- **Our rover design features a four-wheel rocker configuration with a differential linkage, enhancing stability and manoeuvrability, especially in challenging terrains like M sand.**

In short, our rover project brings together advanced tech, careful planning, and teamwork for the upcoming competition. We're using the latest gadgets, planning every detail, and working together to make our rover stand out in robotics.

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Serial No.	Software	Version	Purpose	Remarks
01	Robotic Operating System	Humble	Communication framework between different parts of the robot	Compatibility with the VIO algorithm, also LTS till 2028
02	Gazebo	Garden	Simulation for rover and manipulator testing	Tooling and integration with ROS2 is easy
03	CoppeliaSim [12]	4.6.0	Simulation	Integrated development environment for manipulator arm with embedded inverse/forward kinematics algorithms, powerful API, path planning and data recording
05	Ubuntu	22.04(Jammy Jellyfish)	OS for development	Compatible with most of the other software
06	Git	2.34.1	Version control and collaboration among team members	
07	Visual Studio Code	1.8*	Text editor for all code developed for the rover	Code highlighting and auto-completion for multiple languages and tooling available for ROS2 also
07	Matlab	R2023b	Simulation	Used for initial simulations to understand the movement of manipulator's arm using kinematic model
08	Fusion 360		Rover and manipulator arm 3D Design	
09	Solidworks		Rover and manipulator arm 3D Design and mechanical analysis	

Table 4: Software Requirements

Serial No.	Algorithm Name	Purpose	Remark
01	HybVIO	Visual Inertial Odometry for robot localisation and environment mapping	Light on computation, high accuracy and SLAM capabilities. ROS2 Humble implementation is available (open source).
02	DOPE(Deep Object Pose Estimation)	Detection and 6-DoF pose estimation of objects from an RGB camera.	Compatible with the Raspberry Pi Camera Module 3, generalizes better to novel environments, especially extreme lighting conditions.
03	Crater Detection Using LiDAR	Detection of craters	To be created by the team
04	Open Motion Planning Library (OMPL)	For path planning	To be created by the team

Table 5: Algorithms

Serial No.	Hardware Details	Procurement Source	Specifications with Realization Plan	Quantity	Estimated Cost
01	DC geared motor	Market	Voltage (DC):12V Rated RPM (at 12V):10 Gear Reduction: 810K Torque (N-cm): 680-780, Encoded	6	4200
02	DC Motor Controller	Market	Bidirectional control.Control one unipolar/bipolar stepper motor.Operating Voltage: DC 4V - 16V Maximum Motor Current: 3A - 5A .	2	1500
03	11.1v 8Ah Li-ion battery	Market	High energy density, High load capacity	3	15000
04	3D printing filament (TPU /PLA /Nylon, etc.)	Market/3D print	Durable, flexible	5	10000
05	Pixhawk	Market	Ease of use, reliable connectivity and communication	1	17000
06	Electromagnetic Clutch	Market	Ease of control and compatibility	4	6000
06	Aluminium Extrusions and Sheets	Market/Indigenous	Durable, lightweight, Modular	10	3000
07	Steel angles/ bars/ rods	Market/Indigenous	Durable, Stiffness, Strength	2	1000
08	Miscellaneous,Tools	Market	Nut , bolts , Screws , wires , etc.		20000

Table 6: Hardware Realisation for Rover Mechanism

Serial No.	Hardware Details	Procurement Source	Specifications/Realization Plan	Quantity	Estimated Cost
01	Servo Motors	Market	Holding Torque:2Kg.cm - 70kg.cm Operating Voltage: 12V - 14V Precise Magnetic Encoder	5	10000
02	Stepper Motors	Market	Holding Torque: 1.6 kg-cm, Voltage: 12 24VDC. Type: Bi-Polar. Precise Magnetic Encoder	1	5000
03	USB Camera	Market	module size : 4cm*4cm, FOV :80° to90°(H), resolution : 2500 × 1900 to 2600 × 1950	1	5000
04	Pressure Sensor.	Market	Trigger Force: j 20g, triggered (default resistance;200kΩ). Pressure Measuring Range: 20g 10kg or larger	2	900
05	Force Sensing Resistor	Market	Force Sensitivity Range: 0.2N – 20N	2	1400
06	Driver board	Market	20 Amp Driver boards with 2 channels each	4	5100
07	Frame Components	Market/Indigenous	6mm thick acrylic plates	5	6000
08	Battery	Market	10000 mAh Manipulator Battery	1	20000
09	IR sensor.	Market	Distance measuring range: 20 to 150 cm. Analog output type. Consumption current: Typ. 33 mA. Supply voltage: 4.5 to 5.5 V	2	2500
10	Miscellaneous	Market/3D print	Nut , bolts , Screws , wires , etc.		20000

Table 7: Hardware Realisation for Manipulator

Serial No.	Hardware Details	Procurement Source	Specifications/Realization Plan	Quantity	Estimated Cost
01	MID-360 LIVOX	Market	FOV: 360° * 59°; Min. Detection Range: 0.1m; L × W × H (mm): 65*65*60; Weight: 265g;	1	62000
02	Flir Firefly S	Market	Resolution: 1440 x 1080; Megapixels: 1.6; Max Frame Rate Standard: 60 Hz; Pixel Size: 3.45; Sensor Type: CMOS; Data Interface: USB 3.1 Gen 1;	2	39000
03	VectorNav VN-100	Market	32-bit processor; Power: 220 mW; Accelerometer Range: ±16 g; Gyroscope Range: ±2,000° /sec; Pitch/Roll: 0.5°; IMU Data: 800Hz	1	80000
04	NVIDIA@ Jetson AGX Xavier	Market	GPU: NVIDIA Volta architecture with 512 NVIDIA CUDA cores and 64 Tensor cores; CPU: 8-core NVIDIA Carmel Armv8.2 64-bit CPU 8MB L2 + 4MB L3; Storage: 32GB eMMC 5.1	1	99,000

Table 8: Hardware Realisation for Navigation

Component	Specification
Rover Type	Autonomous Ground Rover
Rover chassis Dimensions	~ 60 * 30 * 12cm
Ground clearance	~ 30cm
Wheel dimension	~ 20 * 8cm
Rover Weight	~ 20kg
Driver mechanism	4 wheels
Suspension System	Rocker Bogie with Differential Linkage
Power Source	Li-ion battery
Power Consumption	~ 75W
Emergency stop mechanism	Kill switch
Communication	RF(Radio Frequency)
Camera	FLIR Firefly@ S , 720p USB camera
Li-DAR	MID-360 LIVOX
OS	Ubuntu, ROS
Communication protocol	Serial UART, I2C, CAN, TCP/IP, USB
Algorithms	DOPE, SLAM, HybVIO, OMPL
Control Mechanism	Inverse Kinematics
Links	5
Joints	5
Degree of Freedom	~ 5
Actuators	Servo motors, stepper motor
Payload handling	300gm - 500gm

Table 9: System Specification

Serial NO	Task	Main Responsibility	Deadline for Completion	Estimated Cost	Second Responsibility
1	System Design(CAD Modeling)	Adam Alvares, Nishant Chaudhari	Completed	system cost?	Rahul Kulkarni
2	Computation Cost, Simulation	Sattwik Sahu, Siddhi Lipare,	30/01/24	system cost?	Rugved Upaddye
3	Calculations, finalize hardware models	Adam Alvares, Nishant Chaudhari	15/02/24	0	Abhishek Prasad
4	Design Report	Vinita Sao, Abhishek Prasad	28/02/24	0	Saifullah Khan, Rugved Upaddye
5	Hardware Procurement	Vinita Sao	10/03/24	460000	Abhishek Prasad
6	Prototype(3D Printing)	Rahul Kulkarni, Adam Alvares	5/05/24	20000	Saifullah Khan
7	Prototype(Electronics)	Abhishek Prasad	5/05/24	0	Sumit Pisal
8	Prototype(Chassis building)	Rahul Kulkarni, Nishant Chaudhari	5/05/24	0	Adam Alvares
9	Prototype Testing(rover)	Abhishek Prasad, Sumit Pisal	10/05/24	10000	Sattwik Sahu, Vinita Sao
10	Prototype Testing(Manipulator)	Saifullah Khan, Vinita Sao	10/05/24	0	Nishant Chaudhari
11	Prototype Iteration(rover)	Abhishek Prasad, Sumit Pisal	10/05/24	0	Rahul Kulkarni
12	Prototype Iteration(Manipulator)	Saifullah Khan, Vinita Sao	10/05/24	0	Nishant Chaudhari
13	Real-time Testing(Integration)	Abhishek Prasad, Vinita Sao	01/07/24	0	Saifullah Khan, Sattwik Sahu
14	Real-time Iteration	Abhishek Prasad, Vinita Sao	01/07/24	0	Saifullah Khan, Sattwik Sahu

Table 10: Project Management